



ENGG100 - Practical 4

Engineering Computing and Analysis (University of Wollongong)



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ENGG100 – Engineering Computing and Analysis

Practical 4 – Week 4

In this Practical class, you will work through the last parts of the second chapter of the subject textbook “Essential MATLAB for engineers and scientists” (7th Edition) by Brian H. Hahn and Daniel T. Valentine, 2019.

STEP 1 – Access the online textbook through the library’s portal.

- 1.1 Access the textbook using the same process as Practical 1. Access the PDF version of Chapter 2. Note the objectives of each chapter are highlighted in orange.

STEP 2 – Section 2.8 – Decisions.

- 2.1 Read and perform the tasks in Section 2.8 – Decisions and Section 2.8.1 – The one-line **if** statement. Perform the Exercises and check if there were any discrepancies to what you expected.
- 2.2 Read and perform the tasks in Sections 2.8.2–2.8.7. Now revisit your flowchart and structure plan from **Practical 3, Step 4.2** - for finding the solutions to a quadratic equation. If you need revision, look at page 94 in Chapter 3.1 – The Program Design Process. Implement your flowchart solution with the use of **if / if-else** statements. Compare your solution with the code on Pages 95 to 96.
- 2.3 Read and perform the tasks in Section 2.8.9 – The **switch** statement. Consider how would you modify your program in Step 2.2 using the **switch** statement instead of **if/if-else**? You may want to re-write your program outside of class time in order to practice the use of **switch** statements.

STEP 3 – Section 2.7 – Repeating with for.

- 3.1 Read and perform the tasks in Section 2.7 – Repeating with **for**. This is a general introduction to **for** loops. We can use **for** loops to implement algorithms that require repetitive computation, such as Newton’s method for square roots, factorials and the limit of a sequence. Read through Sections 2.7.1–2.7.3 and select one example to complete.
- 3.2 Read carefully Section 2.7.4 – The basic **for** construct and Section 2.7.5 – **for** in a single line. We will discuss more on vectorisation later in the subject. For now, go to the end of Section 2.7 (page 63) and select one of the series calculations to perform using **for** loops.
- 3.3 For additional practice using repetition, go to the end-of-chapter exercises and answer questions 2.19 and 2.20 on page 78.